

DINGYUN HUANG

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EDUCATION

2021 – 2026	University of Cambridge – PhD in Physics Data-driven methods for materials discovery applications in optoelectronics
2017 – 2021	University of Cambridge – BA & MSci in Natural Sciences 1 st class honours with specializations in Physics
2015 – 2017	Lime House School – A Levels

SCHOLARSHIPS & AWARDS

2025	Royal Society of Chemistry Materials Chemistry Horizon Prize as a part of the team led by Prof. Jacqueline Cole for the development of chemistry-aware artificial intelligence software, its application to data-driven materials discovery, and its provision of open-source materials databases and language models for the global scientific community.
2021 – 2025	Cambridge Trust PhD Scholarship at the University of Cambridge
2017 – 2021	Churchill College Scholarship for outstanding performance in Tripos examinations
2016 – 2017	Top Gold Awards (UK top 50) in the British Physics and Astronomy & Astrophysics Olympiads

RESEARCH PROJECTS

University of Cambridge – Research Associate

2025-2026	Optimization of galvanizing line control with deep learning with Tata Steel UK - Prediction of zinc coating weight with 1% fractional error by originating a deep learning model and a data pipeline to align, clean, and resample production line signals. - Trained a feedback models that restores coating weight deviations under ten time-steps by optimizing production parameters according to the target and achieved coating weights on-the-fly.
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University of Cambridge – PhD in Physics

2024 – 2025	End-to-end workflow of training machine learning predictors for PL wavelengths - Extracted a dataset at above 90% precision of experimental PL wavelengths from the literature. - Acquired 20% more data via a solution for extracting data from Wiley articles PDFs. - Predicted PL wavelengths of TADF molecules at 0.12 eV MAE with a heterogeneous GNN model. - Published results in <i>ACS J. Chem. Inf. Model.</i> doi/10.1021/acs.jcim.6c00425
2023 – 2024	Cost-efficient fine-tuning of large language models for optoelectronics - Pre-trained and fine-tuned three BERT-like language models for optoelectronics tasks, including data-mining through multi-turn Q&A, and text embedding. - Conducted a detailed comparison of the efficiency of domain adaptation technique for BERT and RoBERTa; Showed that models adapted from RoBERTa required an 80% less pre-training than those from BERT to reach similar performance. - Published results in <i>ACS J. Chem. Inf. Model.</i> doi.org/10.1021/acs.jcim.4c02029 cited by 9.
2021 – 2023	Automatic database collection for TADF materials through text-mining - Achieved 82% precision in curating an OLED materials database by devising an automated text-mining pipeline basing on ChemDataExtractor. The database has been downloaded over 800 times and has been used in three peer-reviewed journal articles since its release in 2024. - Improved the chemical assignment precision from 30% to 80% by innovating a statistical screening method for compound name extraction. - Designed an OPSIN-based workflow of 92% accuracy on deducing molecular structure of a compound from journal texts; this package PyOPSIN now has ten thousand annual downloads on PyPI. - Published results in <i>Nature Scientific Data</i> doi.org/10.1038/s41597-023-02897-3 cited by 31.

2021 - present	ChemDataExtractor - Boosted data extraction recall by 30% by locating and fixing multiple critical issues in ChemDataExtractor modules of abbreviation detections, text and table parsing, and XML scraping. - Actively maintained the group software ChemDataExtractor2, resolving issues raised on GitHub. - Successfully migrated ChemDataExtractor from AllenNLP to Torch and HuggingFace and upgraded the software support from Python 3.6 to the Python 3.11. - Cut installation time from hours to a few minutes by systemizing the required dependencies.
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WORK EXPERIENCES

2025 – 2026	Department of Physics, University of Cambridge – Research Associate Predicted zinc coating weight with 1% error during steel sheet galvanization from production parameters by designing a bespoke data pipeline and a deep learning model. Optimized parameters when coating weight deviates from designed values to reduce zinc waste by implementing a counterpart feedback model.
Summer 2025	Huawei Hong Kong Research Centre, Theory Lab – Research Scientist Internship Reduced regression errors by 65% while reducing the number of trainable parameters by 90% by innovating a physics-informed neural networks for optical imaging systems optimizations. Significantly reduced dataset loading time and memory overhead by implementing a sharding methods.
2024 – 2025	Churchill College, University of Cambridge – Mathematics Supervisor Simplified advanced mathematical concepts and improved teaching of problem-solving skills . Fostered a supportive and effective learning environment for six Natural Sciences students in learning Mathematics, five out of six of them achieved first class in their examination.
Apr – Sep 2022	University of Cambridge STEM Smart Project – Supervisor Designed and delivered interactive pre-university mathematics lessons for non-fee-paying six form students, contributing to the Cambridge STEM Smart initiative. Helped bridge knowledge gaps for underrepresented students, boosting their preparedness for subject challenges and higher education.
Jan – Mar 2022	University of Cambridge – Python Computing Demonstrator Introduced numerical methods used in physics and their real-world applications. Improved tailored exercises to address varying skill levels, resulting in a marked improvement in students’ programing proficiency.

TECHNICAL SKILLS

- ▶ **Python:** PyTorch, Scikit-Learn, Pandas, and other popular data-science/ML packages; large-scale data manipulation and visualization; experienced in object-oriented programming and code packaging for managing ChemDataExtractor and group websites.
- ▶ **High performance computing:** exascale supercomputing clusters; MPI-based multi-process workflows; multi-GPU and multi-node distributed model training; website front and back end through AWS.
- ▶ **ML Algorithms:** CNN, GNN, Transformers, LSTM, Random Forests, Bayesian methods, feature engineering.
- ▶ **Language Modelling:** conventional NLP methods; pre-training and fine-tuning transformer-based language models, such as BERT and Llama; deploying language models on downstream applications such as knowledge extraction through multi-turn QA and RAG. Agentic workflow for scientific discoveries.
- ▶ **Material modelling:** cheminformatics toolkit RDKit; Cambridge Structural Database; property modelling using graph neural networks and other ML algorithms.
- ▶ **Git & PyPI:** version management and code collaboration on group software through GitHub; feature unit testing and package publishing via PyPI.
- ▶ **Mathematics:** postgraduate level calculus, tensor algebra, statistics, differential equations, probabilities
- ▶ **Language:** Chinese (native); English (fluent); Cantonese (basic)

INTERESTS & ACTIVITIES

2021 – present	Churchill College Basketball Team (topped the college league for 5 seasons)
2019 and 2022	Oriental Dance Association Treasurer (2022), Publicity and Sponsorship Officer (2019)
2018 – 2019	Cambridge University Chinese Cultural Society Sports Officer